Customer Behaviour & Marketing Campaign Analysis

Excel Data Analytics Portfolio Project | Customer Segmentation & Marketing Performance

Project Overview

How do customer characteristics relate to purchasing behaviour, and what can businesses learn from marketing campaign responses?

This project explores customer purchasing behaviour and campaign engagement using Microsoft Excel. The objective was to identify spending patterns, examine relationships between customer income and expenditure, and explore opportunities for more effective customer segmentation.

The analysis uses an educational dataset adapted from an iFood recruitment case study, containing 2,021 customer records.

Project context: Completed as part of a guided Data Analytics Accelerator (DAA) learning programme.

Key Findings

1. Higher-income customers generally spend more

The analysis identified a positive association between customer income and total spending.

However, substantial variation exists among customers with similar incomes, suggesting income alone is insufficient to explain purchasing behaviour.

Business implication: Combining income with purchasing history and other customer characteristics may provide more useful segmentation than relying on income alone.

2. Campaign 6 respondents have higher average spending

Customer group	Customers	Average spending
All customers	2,021	563.79
Campaign 6 respondents	311	917.26

Customers who responded to Campaign 6 had substantially higher average spending than the overall customer population.

However, this comparison does not establish that the campaign increased spending. Respondents may already have been more valuable or engaged customers before the campaign.

Business implication: Investigate which characteristics predict campaign response and use appropriate control groups to evaluate whether future campaigns generate incremental value.

3. Customer spending is unevenly distributed

The spending distribution is positively skewed, with a relatively small group of high-spending customers and a larger concentration of customers at lower expenditure levels.

Business implication: Businesses could explore customer-value segments and investigate whether different groups respond to different marketing approaches.

Tools & Methodology

Microsoft Excel was used throughout the analysis.

The project involved:

- Data exploration, filtering and descriptive statistics.
- Calculating customer spending and campaign response summaries.
- Using XY scatter plots to examine relationships between income and spending.
- Using histograms to investigate spending distributions.
- Comparing Campaign 6 respondents with the overall customer population.
- Interpreting findings and identifying potential marketing applications.

Business Recommendations

Based on the exploratory findings, three areas warrant further investigation:

1. Develop customer segments: Combine income, purchase history, age and campaign engagement to understand differences in customer behaviour.
2. Assess campaign targeting: Examine which customer groups have higher response rates and whether these groups also demonstrate higher customer value.
3. Evaluate incremental campaign impact: Use randomised control groups or suitable comparison methods to distinguish existing customer differences from genuine campaign effects.

These recommendations are opportunities for further analysis rather than proven improvements in marketing performance.

Limitations

This is an exploratory, observational analysis and does not establish causal relationships.

- Campaign respondents may differ systematically from non-respondents.
- Higher average spending among respondents does not demonstrate campaign effectiveness.
- Income and spending refer to different reporting periods.
- Monetary units in the source dataset have not been independently confirmed.
- The dataset is educational and may not fully represent real-world customer populations.

Repository Contents

- Customer_Campaign_Analysis.xlsx — Excel workbook containing the data, calculations and visualisations.
- Customer_Campaign_Analysis_Overview.pdf — Project summary, findings, recommendations and limitations.

How to Explore the Analysis

1. Download the Excel workbook.
2. Open it in Microsoft Excel.
3. Review the customer analysis worksheets and supporting calculations.
4. Examine the income-versus-spending scatter plots and spending distributions.
5. Compare the Campaign 6 respondent group with the overall population.

Data Source & Project Attribution

Educational customer marketing dataset adapted from the iFood Recruitment Case Study.

Completed as part of a guided Data Analytics Accelerator learning project. The analytical interpretation and documentation have been developed for portfolio presentation with AI assistance.

Project Takeaway

This project demonstrates how relatively straightforward Excel analysis can generate useful commercial questions about customer behaviour.

The principal learning is that identifying a relationship is different from demonstrating its cause. Turning descriptive findings into marketing decisions requires careful interpretation, appropriate segmentation and further testing.